

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Faculty: Dr. V. Parthipan

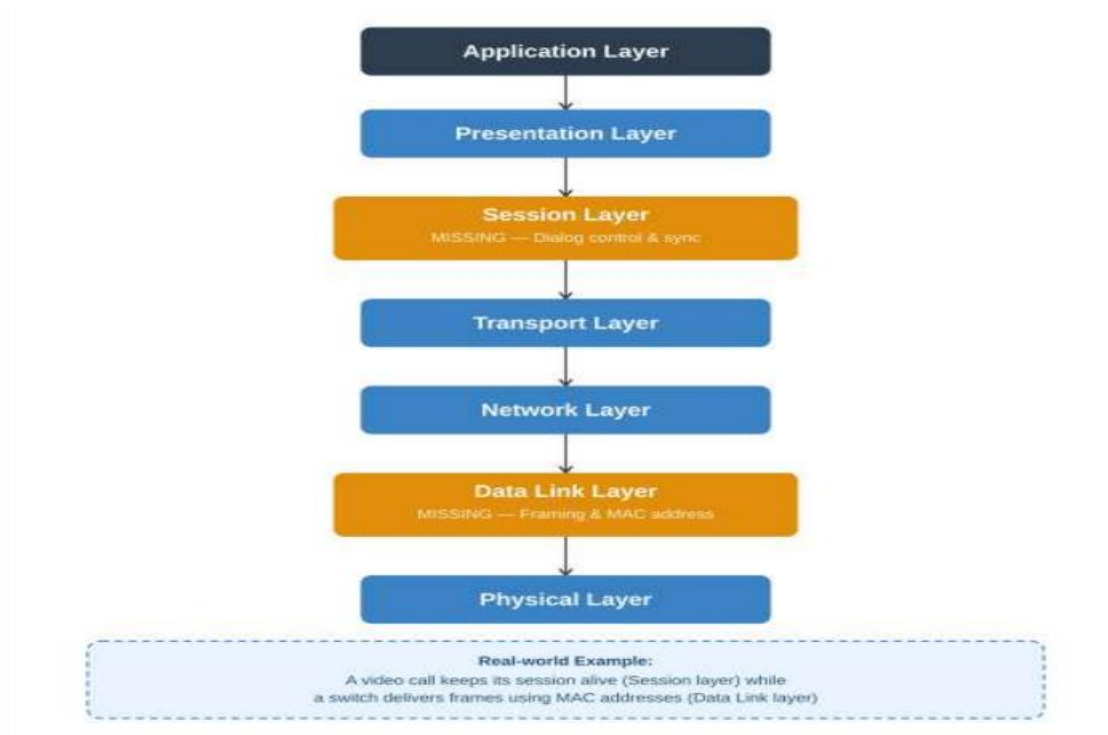
Code of Conduct

I C KARTHIK REDDY(192425157) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C KARTHIK REDDY

Question 1: Missing Layers in the OSI Model

Concept Map:



Analysis of Missing Layers

Session Layer (Layer 5)

The Session Layer is responsible for establishing, managing, and terminating communication sessions between applications. Its key functions include:

- **Dialog Control:** Determines which device transmits at any given time, enabling half-duplex or full-duplex communication
- **Synchronization:** Inserts checkpoints into data streams, allowing recovery from mid-transmission failures without restarting the entire transfer
- **Session Recovery:** Manages reconnection and state restoration if a session is interrupted

Without the Session Layer, applications would lack the mechanism to coordinate multi-request exchanges such as file transfers, login sessions, or database transactions.

Data Link Layer (Layer 2)

The Data Link Layer provides node-to-node delivery of data within the same network segment. Its responsibilities include:

- **Framing:** Organizes raw bits from the Physical Layer into structured frames

- Physical Addressing: Uses MAC addresses to identify source and destination devices on a local network
- Error Detection: Employs techniques like Cyclic Redundancy Check (CRC) to identify corrupted frames
- Flow Control: Regulates data transmission rate between adjacent nodes to prevent buffer overflow

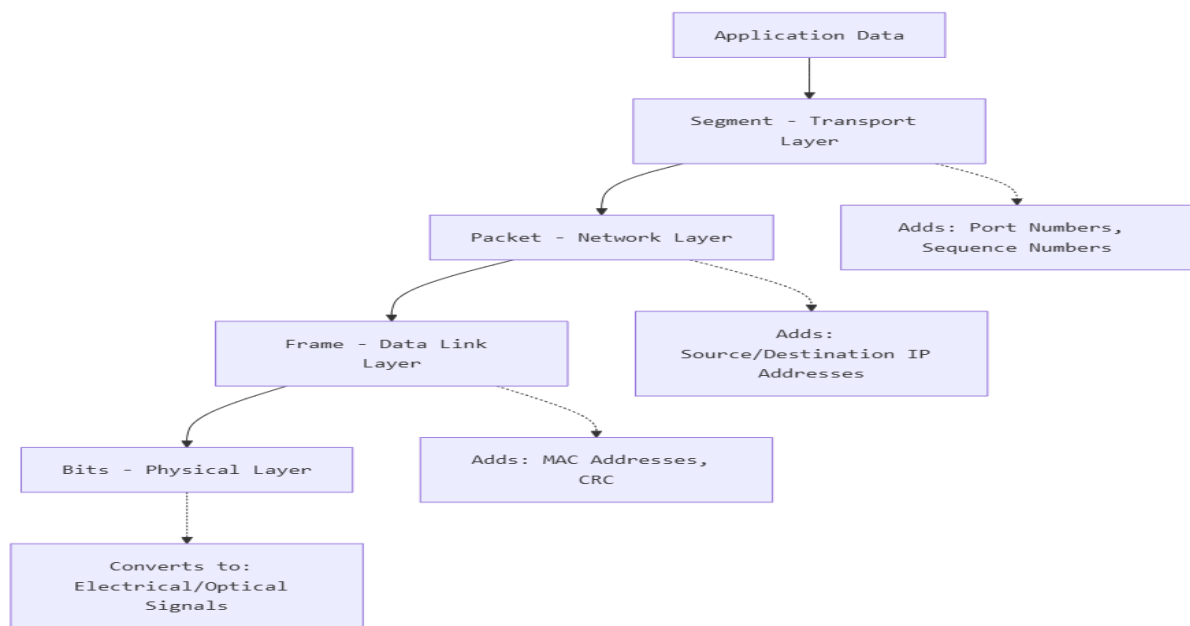
Importance for Reliable Communication

These two layers serve complementary purposes: the Session Layer ensures orderly, synchronized, and recoverable dialogues at the application level, while the Data Link Layer guarantees error-free frame transmission across physical media. Together, they form essential components of the reliability architecture in the OSI model.

Question 2: Data Units and OSI Layers

Concept Map

Data Unit	OSI Layer	Function
Segment	Transport Layer (Layer 4)	Divides application data into manageable units with port numbers
Packet	Network Layer (Layer 3)	Adds logical addressing (IP) for routing across networks
Frame	Data Link Layer (Layer 2)	Encapsulates packets with MAC addresses and error detection
Bits	Physical Layer (Layer 1)	Converts frames into electrical, optical, or radio signals



Application Data

Segment - Transport Layer

Packet - Network Layer

Frame - Data Link Layer

Bits - Physical Layer

Adds: Port Numbers, Sequence Numbers

Adds: Source/Destination IP Addresses

Adds: MAC Addresses, CRC

Converts to: Electrical/Optical Signals

Analysis: How Data Unit Transformation Supports Transmission

Encapsulation Process

As data descends through the OSI stack, each layer performs encapsulation—adding its own header (and sometimes trailer) containing control information specific to that layer's function.

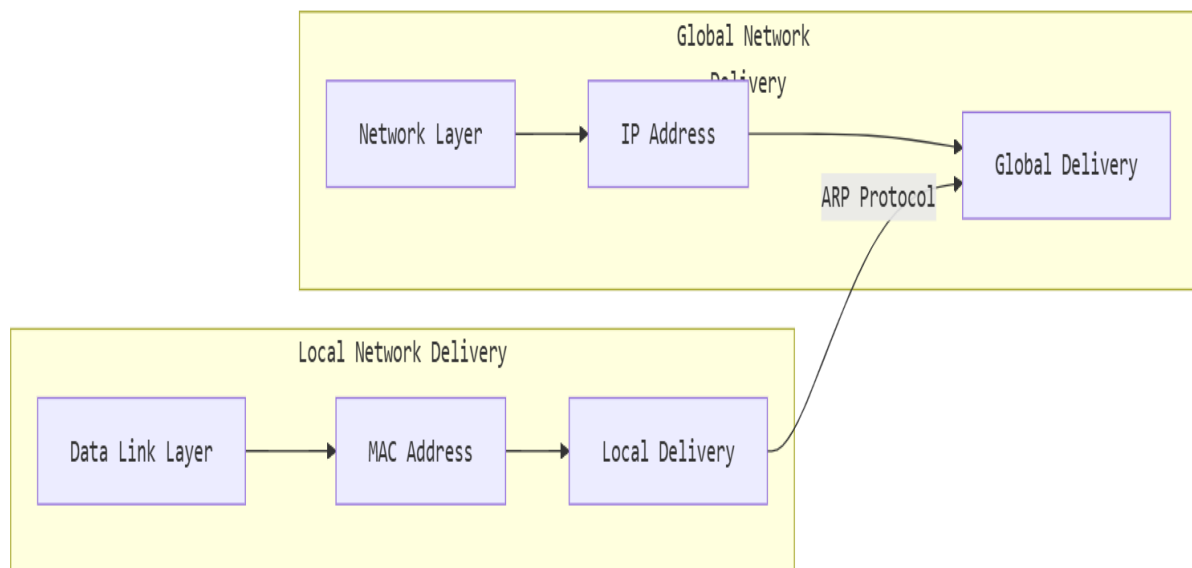
1. **Transport Layer:** The application data is segmented into smaller units. Each segment receives a header containing port numbers (to identify applications), sequence numbers (for ordering), and checksums (for integrity verification).
2. **Network Layer:** Each segment is encapsulated into a packet with source and destination IP addresses added. This logical addressing enables routing decisions across interconnected networks.
3. **Data Link Layer:** Packets are encapsulated into frames with MAC addresses for local delivery and CRC values for error detection at the physical transmission level.
4. **Physical Layer:** Frames are converted into a stream of bits represented as electrical voltages, light pulses, or radio waves appropriate for the transmission medium.

Benefits of Layered Transformation

- **Modularity:** Each layer operates independently, understanding only its own data unit and header
- **Scalability:** Changes to one layer's protocol do not require modifications to other layers
- **Reliability:** Each layer can implement its own error handling and flow control mechanisms
- **Interoperability:** Devices from different manufacturers can communicate as long as they adhere to layer-specific protocols

Question 3: MAC Address and IP Address Relationships

Concept Map



Analysis: Ensuring Complete Data Delivery

MAC Address (Physical Address)

- A 48-bit hardware address burned into the network interface card
- Used exclusively for communication within a local network segment (LAN)
- Switches use MAC addresses to forward frames to the correct port
- Only meaningful within a single broadcast domain

IP Address (Logical Address)

- A 32-bit (IPv4) or 128-bit (IPv6) address assigned through network configuration
- Used for routing packets across multiple interconnected networks
- Routers examine destination IP addresses to determine the optimal path
- Hierarchical structure enables efficient routing table organization

Complementary Operation

These addressing schemes work together through the Address Resolution Protocol (ARP):

1. The source device knows the destination's IP address but needs the MAC address for local delivery
2. ARP broadcasts a request asking "Who has this IP address?"
3. The device with that IP responds with its MAC address
4. The source can now construct a frame with the correct destination MAC address

End-to-End Delivery Example

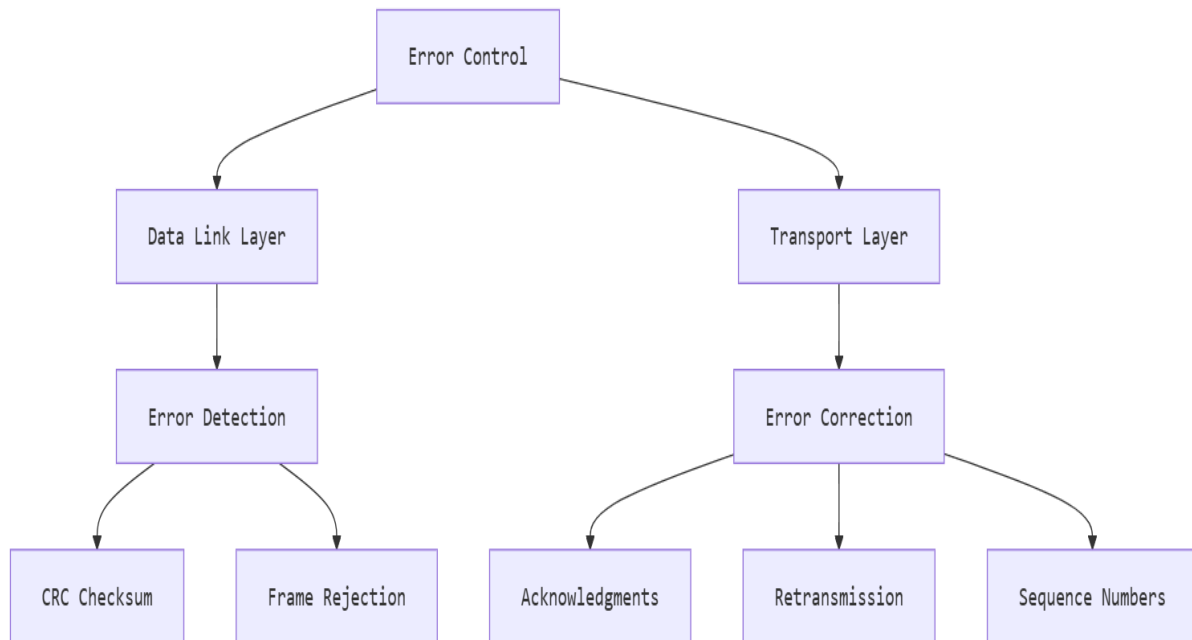
When data travels from Host A to Host B across multiple networks:

- The IP address remains constant throughout the journey, identifying the final destination
- The MAC address changes at each hop: first to the local router, then to the next router, and finally to Host B
- Routers decapsulate frames, examine IP addresses, make routing decisions, then re-encapsulate with new MAC addresses

This dual-addressing system ensures that packets reach the correct network globally (via IP) and the correct device locally (via MAC).

Question 4: Error Control Mechanisms

Concept Map



Analysis: Layered Reliability Mechanisms

Data Link Layer: Error Detection

The Data Link Layer implements error detection at the frame level using:

- **Cyclic Redundancy Check (CRC):** A polynomial-based algorithm that generates a check value appended to each frame
- **Checksum:** A simpler arithmetic sum of frame contents
- **Parity Bits:** Basic single-bit error detection

When a frame arrives, the receiving device recalculates the check value. If it doesn't match the transmitted value, the frame is known to be corrupted and is typically discarded. This layer detects errors close to their source, preventing corrupted data from consuming bandwidth further along the path.

Transport Layer: Error Correction

The Transport Layer (particularly TCP) provides end-to-end reliability through:

- **Sequence Numbers:** Each segment is numbered, allowing the receiver to detect missing or out-of-order segments
- **Acknowledgments (ACKs):** The receiver confirms successful receipt of segments
- **Timers:** If an ACK isn't received within a timeout period, the segment is retransmitted

- Sliding Window: Controls how many segments can be in transit before requiring acknowledgment

Interaction and Reliability Improvement

Layer	Scope	Mechanism	Action on Error
Data Link	Adjacent nodes	CRC, Checksum	Discard corrupted frame
Transport	End-to-end	ACK, Retransmission Request	missing segment

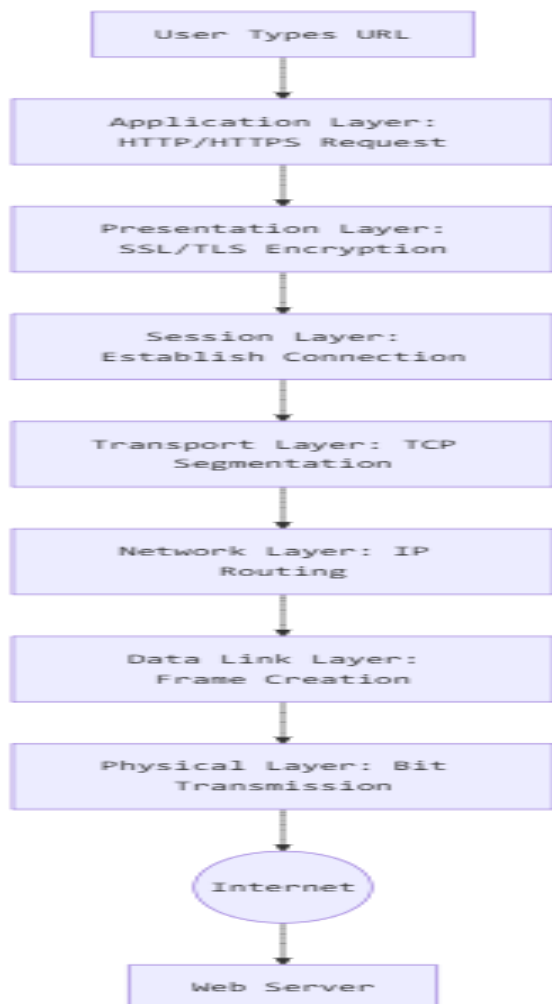
This layered defense creates a robust error control system:

1. The Data Link Layer catches transmission errors quickly and locally, preventing bad data from propagating
2. The Transport Layer provides a safety net for the entire path, ensuring complete and ordered delivery even if intermediate hops fail to catch errors

Neither layer alone would be sufficient—local detection without end-to-end correction would leave gaps, while end-to-end correction without local detection would waste bandwidth transmitting known-bad data.

Question 5: Web Browsing Using OSI Layers

Concept Map



Analysis: Layer Dependencies in Web Browsing

Normal Operation Sequence

1. Application Layer: Browser generates HTTP/HTTPS request for the webpage
2. Presentation Layer: Data is encrypted using SSL/TLS for secure transmission
3. Session Layer: A session is established to maintain connection state
4. Transport Layer: TCP divides data into segments with reliability mechanisms
5. Network Layer: IP addresses are added and routing paths are determined
6. Data Link Layer: Frames are created with MAC addresses for local delivery
7. Physical Layer: Data is converted to electrical/optical signals for transmission

Impact of Layer Failures

Failed Layer Direct Effect		Cascading Impact
Physical	No signal transmission	Complete communication failure
Data Link	Frames cannot reach local gateway	No packets leave local network
Network	Packets cannot be routed	Transport connections fail
Transport	No reliable delivery	Applications cannot exchange data
Session	Connection state lost	Long transfers interrupted
Presentation	Encryption/decryption fails	Secure sites inaccessible
Application	HTTP processing fails	Webpage cannot load

Dependency Analysis

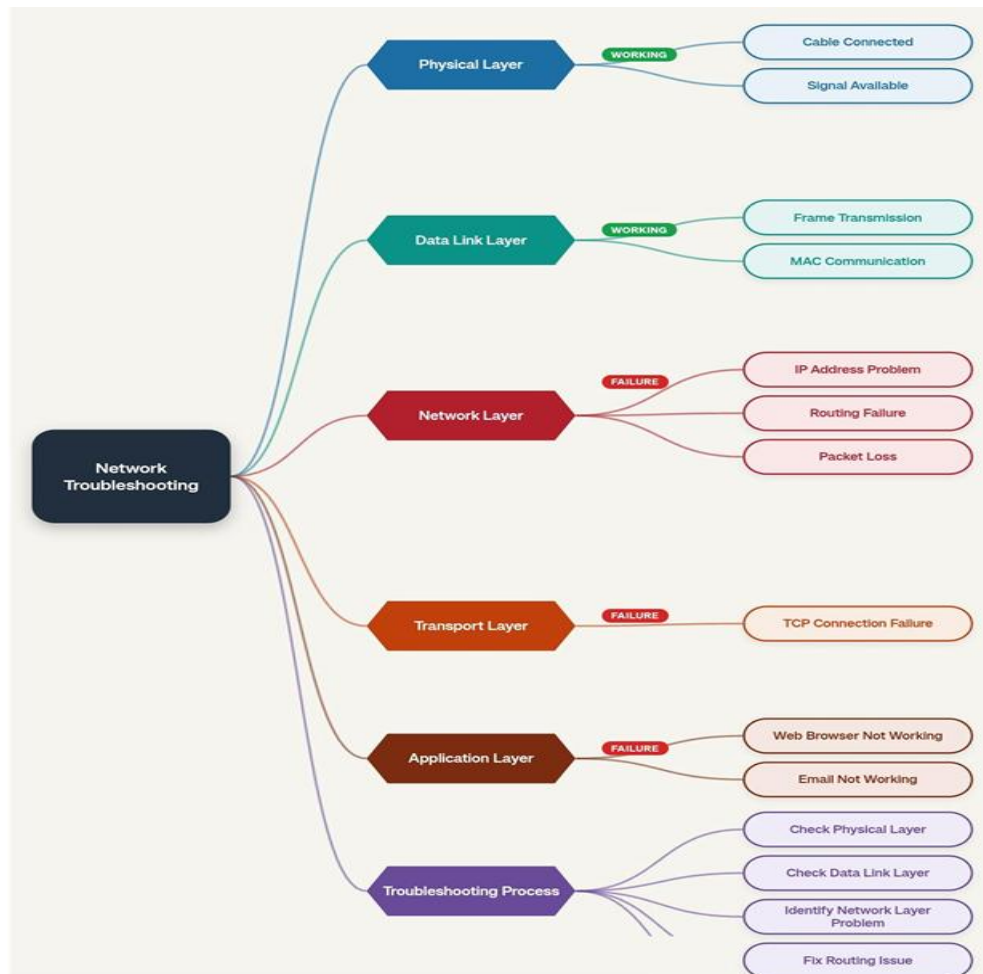
The OSI model exhibits strict hierarchical dependency—each layer relies on all layers below it functioning correctly. A failure at any layer renders all upper layers non-functional, regardless of their individual health:

- A corrupted Physical Layer (broken cable) makes the entire stack useless
- A misconfigured Network Layer (wrong IP settings) prevents Transport Layer connections even with perfect physical connectivity
- A certificate error at the Presentation Layer blocks HTTPS sites even though all lower layers work perfectly

This dependency structure emphasizes why systematic, bottom-up troubleshooting is the most efficient approach to diagnosing network problems.

Question 6: Network Troubleshooting Using Concept Mapping

Concept Map



Problem Analysis

Working Layers (✓)

- Physical Layer: Valid electrical/optical signals are present; cables and hardware are functional
- Data Link Layer: Frames are successfully exchanged with adjacent devices; MAC addressing works; the device can communicate with its local switch or access point

Failed Layer (✗)

- Network Layer: This is the root cause of failure; IP-based communication is not functioning

Cascading Failures

Because the Network Layer has failed:

- Transport Layer cannot establish end-to-end connections (TCP handshakes fail)

- Application Layer cannot exchange data (HTTP requests never reach servers)

Possible Causes of Network Layer Failure

Issue	Description	Diagnostic Command
No IP Address	DHCP failed or static IP not configured	ipconfig / ip addr
Incorrect Subnet Mask	Device cannot determine local vs. remote hosts	ipconfig /all
Wrong Default Gateway	Packets cannot leave local network	route print / ip route
Gateway Unreachable	Router is down or misconfigured	ping [gateway IP]
Routing Table Error	Packets sent to wrong destination	tracert / traceroute
DNS Failure	Names cannot resolve to IP addresses	nslookup
IP Address Conflict	Two devices share the same IP	Check event logs

How Concept Mapping Aids Troubleshooting

Systematic Isolation

The concept map visualization allows network administrators to:

1. Identify the failure boundary: The transition from ✓ to ✗ pinpoints exactly where to focus
2. Eliminate unnecessary checks: Since Physical and Data Link layers work, there's no need to test cables, NICs, or switch ports
3. Direct attention precisely: All diagnostic effort should concentrate on Network Layer configuration

Troubleshooting Procedure

Based on this concept map:

1. Verify IP address assignment: ipconfig (Windows) or ip addr (Linux)
2. Check subnet mask and default gateway configuration
3. Ping the default gateway to test local routing
4. Ping a known external IP (e.g., 8.8.8.8) to test internet routing
5. Test DNS resolution if IP connectivity works but names don't resolve

Efficiency Gains

Without the layer-by-layer visualization, technicians might waste time:

- Replacing cables that work perfectly
- Resetting switches that are functioning correctly
- Reinstalling applications that aren't the source of the problem

The concept map transforms troubleshooting from guesswork into a systematic, efficient process.

Conclusion

The OSI model provides a structured, seven-layer framework for understanding network communication. Through concept mapping, the relationships between layers, addressing schemes, data units, and error control mechanisms become clear and actionable.

Key Insights:

- Each OSI layer serves a distinct purpose and depends on all layers below it
- MAC addresses enable local delivery while IP addresses enable global routing
- Error control operates at multiple layers, combining local detection with end-to-end correction
- Data unit transformation through encapsulation enables modular, scalable network design
- Layer-based concept mapping dramatically improves troubleshooting efficiency

Understanding these concepts is essential for designing reliable networks, diagnosing problems quickly, and ensuring efficient data transmission across any network infrastructure.

Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Clear, logical, well-organized map with proper hierarchy	Mostly clear with minor structural issues	Lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking & Relationships	20%	Correctly establishes relationships between layers, functions, and processes	Mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts with strong reasoning	Correct but lacks depth	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Clear, concise, well-structured explanation	Understandable with minor gaps	Unclear or incomplete	No clear or meaningful explanation